

Reflect Points Across Axes

Lesson 9-7

Name: _____

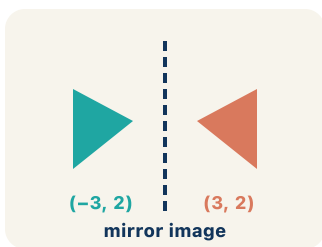
Date: _____

Class: _____

Key Vocabulary

Level 1 support

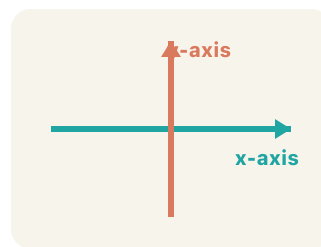
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$(3, 2)$ reflected over the y -axis becomes $(-3, 2)$ — x changes sign, y stays

Reflection

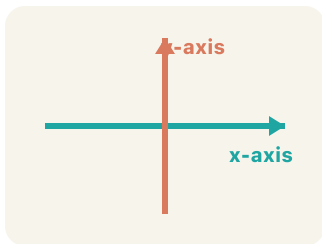
A flipped image of a point on the other side of a line, the same distance away.



$(4, 3) \rightarrow (4, -3)$: the y flips from $+3$ to -3 , like folding the paper along the horizontal line

x-axis

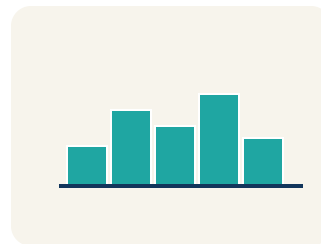
The line that goes across the grid. Flipping over it changes the y sign.



$(4, 3) \rightarrow (-4, 3)$: the x flips from $+4$ to -4 , like folding the paper along the vertical line

y-axis

The line that goes up the grid. Flipping over it changes the x sign.



A butterfly's wings are symmetric — fold it down the middle and both sides match

Symmetry

When a shape looks the same on both sides of a line.

...-2, -1, 0, 1, 2...



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can reflect points across the x-axis and y-axis on the coordinate plane.

1 Notice

Captain Vega discovers that her treasure map has a magic mirror trick!

2 Key idea

I can reflect points across the x-axis and y-axis on the coordinate plane.

3 Words I'll use

Reflection

x-axis

y-axis

Symmetry

Integer



Think About It

- When $(2, 4)$ reflects over the x-axis to $(2, -4)$, which coordinate changed?
- When $(5, 3)$ reflects over the y-axis to $(-5, 3)$, which coordinate changed?
- What stays the same in each reflection?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What is the reflection of $(4, -3)$ over the x-axis?

- A. $(-4, -3)$
- B. $(4, 3)$
- C. $(-4, 3)$
- D. $(3, -4)$

Step 1 Reflecting over the x-axis changes the sign of the y-coordinate.

Step 2 The x stays the same (4), and the y changes from -3 to 3.

Step 3 The answer is $(4, 3)$.



Answer: B. $(4, 3)$

 We do — together

Solve this one with your class using the same steps.

Problem: What is the reflection of $(-2, 6)$ over the y-axis?

- A. $(2, 6)$
- B. $(-2, -6)$
- C. $(2, -6)$
- D. $(6, -2)$

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which point is on the y-axis?

- A. $(0, 5)$
- B. $(5, 0)$
- C. $(3, 3)$
- D. $(-2, 4)$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

When you reflect the point (3, 2) over the y-axis, which coordinate changes its sign and which stays the same? Why?

Level 1 support

Start here: Reflecting over the y-axis changes the ____ coordinate because ____.

 **Need a hint?**

Sentence starters (optional)

The ____ coordinate changes sign.

La coordenada ____ cambia de signo.

The ____ coordinate stays the same.

La coordenada ____ se queda igual.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Level 2

Push students to explain why the reflected point is the same distance from the y-axis as the original.

- The reflected point is the same distance from the axis because ____.
- A reflection is like a ____ over the line.

EXPLORE

Reflect $(4, -5)$ over the x -axis. Talk through how the coordinates change and where the new point lands.

Level 1 support

Start here: Reflecting over the x -axis changes the ____ coordinate, so the point becomes ____.

 **Need a hint?**

Sentence starters (optional)

Over the x -axis, the ____ coordinate flips sign.

Sobre el eje x , la coordenada ____ cambia de signo.

So $(4, -5)$ becomes ____.

Entonces $(4, -5)$ se convierte en ____.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student flips the sign of the y -coordinate, so $(4, -5)$ becomes $(4, 5)$, and explains the x stays the same.

Level 2

Ask students to predict where $(4, -5)$ lands if reflected over the y -axis instead, and compare the two results.

- Reflected over the y -axis, $(4, -5)$ becomes ____.
- The two reflections are different because ____.

CONNECT

How is reflecting a point over an axis like looking at it in a mirror placed on that axis?

Level 1 support

Start here: A reflection is like a mirror because ____.



Need a hint?

Sentence starters (optional)

The axis acts like the ____.

El eje actua como el ____.

The image is the same distance away but on the ____ side.

La imagen esta a la misma distancia pero del ____ lado.

Word bank:

reflection

x-axis

y-axis

coordinate plane

ordered pair

Listen for: Student connects the axis to a mirror line and the reflected point to a mirror image, equal distance on the opposite side.

Level 2

Push students to generalize a rule: what changes when you reflect over the x-axis versus the y-axis?

- Reflecting over the x-axis changes ____, while reflecting over the y-axis changes ____.
- A rule I can always use is ____.

REFLECT

A classmate reflected $(2, 3)$ over the x -axis and got $(-2, 3)$. What did they do wrong, and what is the correct image?

Level 1 support

Start here: The mistake was changing the ____ coordinate, but the correct image is ____.



Need a hint?

Sentence starters (optional)

Over the x -axis, they should have changed the ____ coordinate.

Sobre el eje x , debieron cambiar la coordenada ____.

The correct image is ____.

La imagen correcta es ____.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student identifies they flipped x instead of y ; the correct image over the x -axis is $(2, -3)$.

Level 2

Ask students to critique how to remember which coordinate changes for each axis.

- To remember which coordinate changes, I think ____.
- Reflecting over the x -axis affects the ____ value because ____.

Write About the Math

The Writing Revolution

I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

1. Kernel Sentence subject + verb

Model: Reflection is a flipped image of a point on the other side of a line, the same distance away.
Reflexión es una imagen volteada de un punto al otro lado de una línea, a la misma distancia.

Write a kernel sentence about reflection. Use a subject and a verb.

Escribe una oración base sobre reflexión. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Reflection matters in math
Reflexión importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Reflection matters in math because ____.
Reflexión importa en matemáticas porque ____.

but
pero

Reflection matters in math, but ____.
Reflexión importa en matemáticas, pero ____.

so
entonces

Reflection matters in math, so ____.
Reflexión importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about reflection.
Di un hecho verdadero sobre reflection.

Reflection ____.

Question
Pregunta

Ask a question about reflection.
Haz una pregunta sobre reflection.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about reflection.
Muestra entusiasmo sobre reflection.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with reflection.
Dile a un compañero qué hacer con reflection.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The ____ **coordinate changes sign.**

La coordenada ____ *cambia de signo.*

The ____ **coordinate stays the same.**

La coordenada ____ *se queda igual.*

Over the x-axis, the ____ **coordinate flips sign.**

Sobre el eje x, la coordenada ____ *cambia de signo.*

Try It

Solve on your own. Check the answer key when you are done.

1. A point (6, -1) is reflected across the x-axis. Which coordinate changes sign?

- A. The y-coordinate
- B. The x-coordinate
- C. Both coordinates
- D. Neither coordinate

Show your work:

2. Reflect (-4, -7) across the y-axis. What is the image?

- A. (4, -7)
- B. (-4, 7)
- C. (4, 7)
- D. (-7, -4)

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A triangle has vertices at $A(1, 2)$, $B(4, 2)$, and $C(4, 6)$. If you reflect the entire triangle over the y -axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

Sentence starter: The reflected vertices are $A'(\rule{1cm}{0.4pt})$, $B'(\rule{1cm}{0.4pt})$, $C'(\rule{1cm}{0.4pt})$. I found each by $\rule{1cm}{0.4pt}$. The reflected triangle is $\rule{1cm}{0.4pt}$ compared to the original because reflections $\rule{1cm}{0.4pt}$.

Show your work:

Reflect — Exit Ticket

What is the reflection of $(-5, 2)$ over the y -axis?

- A. $(-5, -2)$
- B. $(5, -2)$
- C. $(5, 2)$
- D. $(2, -5)$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. (2, 6) — *Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2, and y stays 6. The answer is (2, 6).*
2. **You Try (notes):** A. (0, 5) — *Points on the y-axis have x-coordinate = 0. (0, 5) is on the y-axis.*
3. **Try It 1:** A. The y-coordinate — *Reflecting across the x-axis flips the sign of y only, giving (6, 1).*
4. **Try It 2:** A. (4, -7) — *Reflecting across the y-axis changes the sign of x: (4, -7).*
5. **Exit Ticket:** C. (5, 2) — *Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -5 to 5, and y stays 2. The reflection is (5, 2).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Reflection is a flipped image of a point on the other side of a line, the same distance away.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).